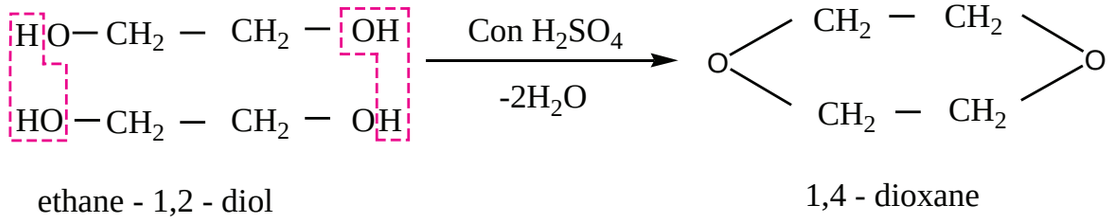
3. When distilled with Conc. H_2SO_4, glycol forms dioxane



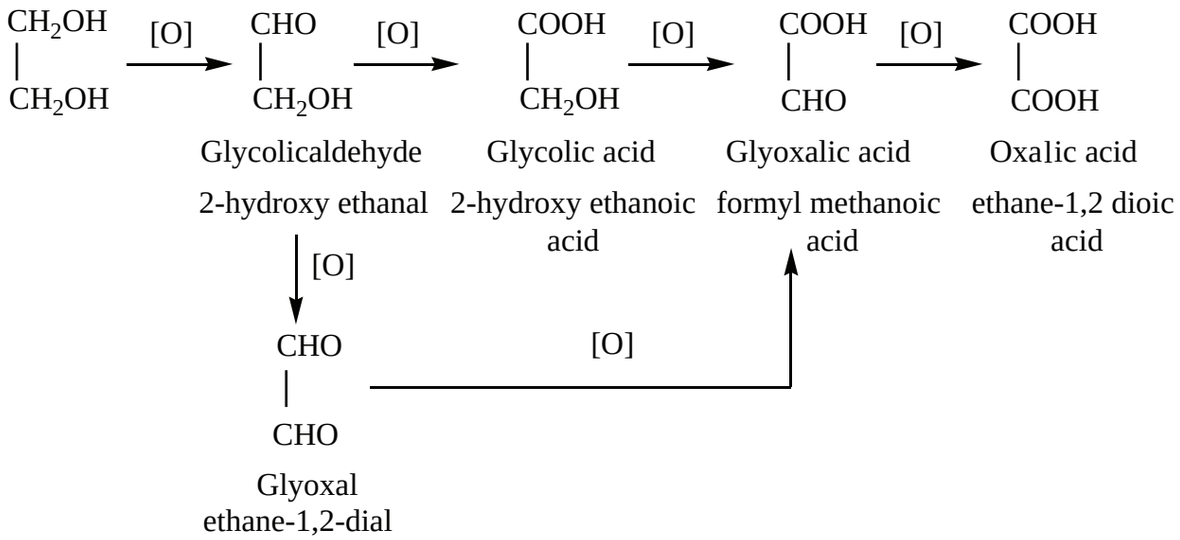
ethane - 1,2 - diol

1,4 - dioxane

Oxidation of glycol

On oxidation, glycol gives a variety of products depending on the nature of oxidizing agent and other reaction conditions.

i) When nitric acid (or) alkaline potassium permanganate is used as the oxidizing agent, the following products are obtained.



Glycolicaldehyde
2-hydroxy ethanal

Glycolic acid
2-hydroxy ethanoic acid

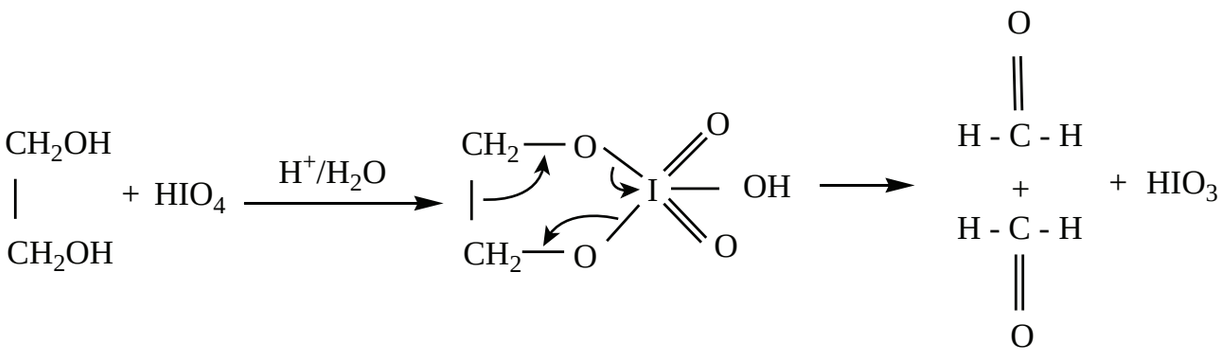
Glyoxalic acid
formyl methanoic acid

Oxalic acid
ethane-1,2 dioic acid

Glyoxal
ethane-1,2-dial

ii) **Oxidation of glycol with periodic acid**

Ethylene glycol on treatment with periodic acid gives formaldehyde. This reaction is selective for vicinal 1,2 – diols and it proceeds through a cyclic periodate ester intermediate.



Formaldehyde

XII U11-Hydroxy compounds.indd 120 11-01-2021 15:44:50